

Course Schedule

Computing, Robotics and Bionics

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Sant Cugat del Vallès, 7-Sep-2026

Course Program

Block A (Sept) – Refreshers & Tools

- Session 1 – Intro to Course, Grading & Setup
- Session 2 – Python & NumPy Fundamentals
- Session 3 – Pandas & Matplotlib
- Session 4 – Calculus and Linear Algebra Refreshers
- Session 5 – Calculus and Linear Algebra Refreshers
- Session 6 – Image Processing (OpenCV)
- Session 7 – LLMs – Theory & AI Coding Assistants

Block B (Oct–Nov) – Classical ML (Géron Part I)

- Session 8 – ML Landscape + E2E Project
- Session 9 – ML Landscape + E2E Project
- Session 10 – Classification
- Session 11 – Training Models
- Session 12 – Decision Trees
- Session 13 – Ensemble Learning
- Session 14 – Dimensionality Reduction
- Session 15 – Unsupervised Learning
- **Session 16 – Miniproject 1 – Presentations**
- **Session 17 – Checkpoint 1**

Block C (Nov) – Neural Networks (Géron Part II)

- Session 18 – Intro to ANNs
- Session 19 – Neural Networks with PyTorch
- Session 20 – Training Deep Neural Networks
- Session 21 – Reinforcement Learning

Block D (Nov–Dec) – Control Theory

- Session 22 – Modelling & Feedback
- Session 23 – PID & Frequency Response
- Session 24 – State-Space & Digital Control

Block E (Dec) – Robotics & Bionics

- Session 25 – Anatomy & Kinematics
- Session 26 – Inverse Kinematics & Motion Planning
- Session 27 – Actuation, Sensing & Bionic Integration
- **Session 28 – Miniproject 2 – Presentations**

Final Exam: date to be confirmed

Grading

The final grade of the subject will be obtained as:

$$\text{Grade} = \text{MAX} (0.7 \cdot \text{Nef}, 0.3 \cdot \text{Nexam1} + 0.4 \cdot \text{Nef}) + 0.3 \cdot \text{Nprojs}$$

Where:

- Nexam1: Checkpoint exam grade
- Nprojs: Projects grade
- Nef : Final exam grade

Exam format:

- Theory (1 h) – Only provided “cheatsheet” (classification performance measures). No materials allowed. Two parts: (1) Kahoot-like MC questions and (2) Open questions from list
- Practice (2 h) – “Open book” (i.e. computer, including internet connection, ChatGPT, etc. allowed)
- Weight: 50% each.

Sample Open Questions

- Supervised vs. Unsupervised Learning
- Regression vs. Classification
- Train/Test
- Stratified Sampling
- Cross Validation
- Performance of an algorithm
- Grid Search
- Linear Regression
- Logistic Regression
- Performance Measures in Classification
- Bayes' Theorem
- XXX

September 2026

Mon	Tue	Wed	Thu	Fri
	1	2	3	4
7 Session 1 Intro to Course, Grading & Setup	8	9	10 Session 2 Python & NumPy Fundamentals	11
14 Session 3 Pandas & Matplotlib	15	16	17 Session 4 Calculus and Linear Algebra Refreshers	18
21 Session 5 Calculus and Linear Algebra Refreshers	22	23	24	25
28 Session 6 Image Processing (OpenCV)	29	30		

October 2026

Mon	Tue	Wed	Thu	Fri
			1 Session 7 LLMs – Theory & AI Coding Assistants	2
5 Session 8 ML Landscape + E2E Project	6	7	8 Session 9 ML Landscape + E2E Project Miniproject 1 released	9
12	13	14	15 Session 10 Classification	16
19 Session 11 Training Models	20	21	22 Session 12 Decision Trees	23
26 Session 13 Ensemble Learning	27	28	29 Session 14 Dimensionality Reduction	30

November 2026

Mon	Tue	Wed	Thu	Fri
2 Session 15 Unsupervised Learning	3	4	5 Session 16 Miniproject 1 – Presentations	6
9 Session 17 Checkpoint 1	10	11	12 Session 18 Intro to ANNs Miniproject 2 released	13
16 Session 19 Neural Networks with PyTorch	17	18	19 Session 20 Training Deep Neural Networks	20
23 Session 21 Reinforcement Learning	24	25	26 Session 22 Modelling & Feedback	27
30 Session 23 PID & Frequency Response				

December 2026

Mon	Tue	Wed	Thu	Fri
	1	2	3 Session 24 State-Space & Digital Control	4
7	8	9	10 Session 25 Anatomy & Kinematics	11
14 Session 26 Inverse Kinematics & Motion Planning	15	16	17 Session 27 Actuation, Sensing & Bionic Integration	18
21 Session 28 Miniproject 2 – Presentations	22	23	24	25
28	29	30	31	

January 2027

Mon	Tue	Wed	Thu	Fri
				1
4	5	6	7	8
11	12	13	14	15
18	19	20	21	22
25	26	27	28	29

Final Exam: date to be confirmed

Assignments

The projects will be executed by teams of 2-3 students. There will be kickoff sessions for each assignment and then 4 weeks to finish the job.

The deliverables shall be Python notebooks executable from the course's Google Colab framework + 1 ppt slide with the following sections: (1) Project description, (2) Approach, (3) Results, (4) Conclusions (template provided) for each one of two different audiences: (1) technical (e.g. colleagues), (2) business (e.g. C-suite).

Each team will present their approach/results during a dedicated 15' session with the help of the Python notebook + 1+1 ppt slides. 10' shall be allocated to the presentation and 5' for Q&A. The presentation shall take place in English.

For each miniproject, "default" problem statements (and template Python notebooks) will be provided, but students are welcome (encouraged!) to come up with their own project proposals (subject to covering the topics of the assignment). Such proposals need to be communicated to the instructor 1 week before the miniproject kickoff session.

Notes:

- You may (should!) use ChatGPT/Gemini/... to enhance your productivity during the miniprojects
- The theory exam may include questions about your assignment (and you won't have ChatGPT or Gemini to help)
- So use ChatGPT/Gemini/... but spend the time to understand (in general) what they are doing.
- ChatGPT/Gemini are also very useful to help come up with exam questions that cannot be answered by ChatGPT/Gemini.

#	Topics	Default	KO Date	Deadline	Presenta tion
1	TBD	TBD	08-10	05-11	05-11
2	TBD	TBD	12-11	21-12	21-12

Miniproject Presentation Slide Template

Presentation to: <Audience>[Technical/C-suite]

Project Description

<Provide a description of the project scope (objectives, specs,...)>

Approach

<What approach did you take>

Results

<Describe the results. One picture is worth a thousand words.>

Conclusions

<Conclusions and next steps.>

Bibliography

McKinney, W., "Python for Data Analysis", 3rd Edition.

Textbook available at <https://wesmckinney.com/book/> - Code files available at <https://github.com/wesm/pydata-book/tree/3rd-edition>



Johansson, R. "Numerical Python", 3rd Edition. Code files available at <https://irjohansson.github.io/numericalpython.html>



Aurélien Géron, "Hands-On Machine Learning with Scikit-Learn and PyTorch"

<https://learning.oreilly.com/library/view/hands-on-machine-learning/9798341607972/> - Code files available at <https://github.com/ageron/handson-ml2>



VanderPlas, J., "Python Data Science Handbook" - <https://www.oreilly.com/library/view/python-data-science/9781491912126/>

Lee, Wei-Meng, "Python Machine Learning" - <https://www.oreilly.com/library/view/python-machine-learning/9781119545637/>

Deisenroth et al., "Mathematics for Machine Learning" - Textbook available at: <https://mml-book.github.io/book/mml-book.pdf>



Lynch, Kevin M., Park, Frank C., "Modern Robotics. Mechanics, Planning, and Control"

Textbook and materials available at https://hades.mech.northwestern.edu/index.php/Modern_Robotics



Bruce, P. et al., "Practical Statistics for Data Scientists", 2nd Ed. - <https://www.oreilly.com/library/view/practical-statistics-for/9781492072935/>

James, G., Witten, D., Hastie, T., Tibshirani, R., Taylor, J. "An Introduction to Statistical Learning with Applications in Python" - Textbook available at <https://www.statlearning.com/>



Hastie, T., Tibshirani, R., Friedman, J. "The Elements of Statistical Learning: Data Mining, Inference, and Prediction" - Textbook and materials available at <https://hastie.su.domains/ElemStatLearn/>



Abu-Mostafa, Yaser S., Magdon-Ismael, Malik, Lin, Hsuan-Tien, "Learning from Data, A Short Course" - Materials available at <https://amlbook.com/>

Goodfellow, I., Bengio, Y., Courville, A., "Deep Learning" - Textbook available at <https://www.deeplearningbook.org/>



Gallier, J., Quaintance, J., "Algebra, Topology, Differential Calculus, and Optimization Theory For Computer Science and Machine Learning" <https://www.cis.upenn.edu/~jean/math-deep.pdf>

Online Resources

Jeremy Balka – Statistics course

<https://www.jbstatistics.com/>



edX – StanfordOnline – Statistical Learning with Python

<https://www.edx.org/learn/python/stanford-university-statistical-learning-with-python>



Andrew Ng's Stanford CS229

<https://www.youtube.com/playlist?list=PLoROMvovdv4rMiGQp3WXShtMGgzqpfVfbU>



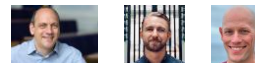
Andrew Ng's "Original" Coursera – Stanford Online - Machine Learning (Matlab/Octave)

<https://www.coursera.org/learn/machine-learning-course/>



Coursera – Imperial College London – Mathematics for Machine Learning Specialization (3 courses)

<https://www.coursera.org/specializations/mathematics-machine-learning>



Coursera Machine Learning Specialization

<https://www.coursera.org/specializations/machine-learning-introduction>



Building a Recommendation System with Python Machine Learning and AI

<https://www.linkedin.com/learning/building-a-recommendation-system-with-python-machine-learning-and-ai>

Grant Sanderson's "3blue1brown"

<https://www.3blue1brown.com/>

Coursera – AI Python for Beginners

<https://www.coursera.org/learn/ai-python-for-beginners>

Modern Robotics: Mechanics, Planning, and Control Specialization

<https://www.coursera.org/specializations/modernrobotics>



Data Sets

https://en.wikipedia.org/wiki/List_of_datasets_for_machine-learning_research

<https://dataportals.org/>

<https://project.opendatamonitor.eu/>

<https://www.openml.org/>

<https://www.kaggle.com/datasets>

<https://archive.ics.uci.edu/>

<https://registry.opendata.aws/>

<https://www.tensorflow.org/datasets>

End of Course Schedule

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Sant Cugat del Vallès, 7-Sep-2026